HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented customer churn prediction models using **Python** (scikit-learn, pandas) and **SQL** against Snowflake data warehouse, identifying key behavioural indicators that reduced predicted churn within targeted cohorts by **18%**.
- Developed and deployed propensity models for cross-sell and upsell opportunities, generating prioritised lists for commercial teams that increased engagement rates by **12%** across targeted accounts.
- Engineered data pipelines in **Python** and **SQL** to extract, transform, and load revenue data from various sources into Snowflake, supporting the creation of attribution frameworks across marketing and sales touchpoints.
- Designed and analysed **A/B tests** for commercial initiatives, providing statistical insights that informed strategic decisions and improved campaign conversion rates by an average of **9%**.
- Owned the end-to-end lifecycle of machine learning models, from feature engineering and training to deployment on **AWS Lambda** and ongoing monitoring, ensuring models remained performant in production.
- Translated ambiguous commercial questions from Sales, Marketing, and Finance stakeholders into well-scoped analytical problems, communicating complex findings clearly to non-technical audiences.
- Containerised machine learning inference services using **Docker** and deployed them to **AWS ECS**, ensuring scalable and reliable model serving for production workloads.

SKILLS

Data Science & ML – Churn Modelling, Propensity Modelling, Next-Best-Action, Revenue Attribution, Experimentation, Causal Inference, Statistical Analysis, Machine Learning, Model Deployment, Feature Engineering, Model Retraining

Programming & Scripting – Python, SQL, pandas, scikit-learn, statsmodels, NumPy

Data Warehousing & ETL – Snowflake, BigQuery, Redshift, Databricks, ETL, Data Pipelines, Power Query

Cloud & Deployment – AWS, SageMaker, Lambda, ECS, EC2, S3, CloudWatch, Docker, Kubernetes

Workflow Automation – n8n, Airflow, Prefect, Dagster

Business Intelligence & Analytics – Power BI, Tableau, DAX, R, RStudio, Data Visualisation

PROJECTS

Customer Churn Prediction & Retention Analysis

- Developed a **churn prediction model** using **Python** (scikit-learn) on a dataset of **250,000 customer records**, achieving an **AUC of 0.82** by identifying key behavioural and transactional features.
- Engineered features from customer interaction logs and transaction history using **SQL** and **pandas**, creating a comprehensive dataset for model training and analysis.
- Collaborated with a cross-functional team to translate model insights into actionable retention playbooks for the Customer Success department, aiming to reduce at-risk customer churn.

Cross-Sell Propensity Modelling for SaaS Products

- Built a **propensity model** in **Python** leveraging user engagement data and product usage patterns to predict likelihood of adopting new features, reaching **75% precision** for high-propensity customers.
- Designed and implemented data extraction pipelines using **SQL** to gather relevant customer data from a cloud data warehouse, feeding into the model training process.
- Developed a **recommendation engine** that suggested next-best-actions to sales representatives, integrated into their CRM workflow to prioritise customer conversations.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)